

About The Project

This project is a flight tracking database management system built using Python with a PyQt5-based GUI and a MySQL database backend.

It uses the pymysql library to connect the interface to stored procedures and views in the MySQL database, allowing users to add airplanes, assign pilots, offer flights, and simulate operations all through an interactive interface.

I. Instructions To Setup App

Install Programming Languages

- Python (most recent version)

Install Packages

- PyQt5 (most recent version)
- pymysql (most recent version)

- 1) Open the terminal in Visual Studio Code (preferred code editor) using Ctrl + backtick
- 2) Ensure the most recent version of Python is installed on your system
- 3) Install the required Python packages: PyQt5 and pymysql
- 4) Start your MySQL server and create a database named flight_tracking
- 5) Add all stored procedures from Phases II and III into the database
- 6) Make sure the GUL.py file is in your project folder

II. Instructions To Run App

- 1) Open your project folder in Visual Studio Code
- 2) Open the terminal if it's not already open
- 3) Confirm that your MySQL server is still running and the database is set up
- 4) Run the Python application from the terminal
- 5) The PyQt5 GUI window should launch, allowing full interaction with the system
- 6) Navigate the GUI menu buttons to find the desired procedure or view

III. Technologies Used & How It Was Created

Listed below are the names and brief descriptions of the primary technologies used in creating the project.

Python

- The entire application is built using the most recent version of Python as the primary language.

PyQt5

- PyQt5 is a Python package used to create the interface that users interact with.

pymysql

- We used pymysql to connect our Python code to the MySQL database.

MySQL

- MySQL is the database management system that stores all the flight and user data, along with the code needed for the procedures and views.

The entire project was developed using Visual Studio Code (VS Code), a widely used code editor. The graphical user interface (GUI) was implemented within a single file, with the PyQt5 and pymysql components based primarily on documentation from CS 2316 and some documentation from the internet. Within the GUI code, each procedure and view was defined in a separate class, which was then turned into individual interactive pages for the user further on in the code. Finally, the pymysql code within each procedure and views class connects to a MySQL database created in Phase III of the project.

IV. Work Distribution

For this project, the work was evenly distributed among the four group members. Below is the specific work distribution.

James Ashton Jones

- Primary creator of the GUI Python code
- Assisted in writing the Read Me

Sarakshi Kandel

- Primary creator of the Read Me document
- Assisted in debugging the GUI file
- Tested the functionality of the GUI file

Vishesh Kathuria

- Assisted in debugging the MySQL file
- Assisted in debugging the GUI file

Viraj Amit Kulkarni

- Assisted in debugging the MySQL file
- Assisted in debugging the GUI file